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Ex: 7 Flow control at Data link Layer using sliding window protocol.

Aim:

To write a program to implement flow control at the data link layer using the sliding window protocol, and to simulate the flow of frames from one node to another.

Algorithm:

1. Start the program
2. Read the total number of frame to be sent (N), the sender's window size (W) and (optionally) a frame number to simulate as lost ~~during~~ during transmission.
3. Initialize base = 0 (sequence number of the oldest unacknowledged frame) and next seq = 0
4. Repeat while base < N:
   a. Determine the current sender window as the range [base, base + W - 1]
   b. Send all frames with sequence number from nextseq up to (base + W - 1), incrementing next.
   c) If the lost frames lies within the frames just sent:
      i) The receiver discards the lost frame and only subsequence out-of-order frames.

d) Else (no loss in this batch)

5. Repeat step 4 until all N frames have been transmitted and acknowledge.
6. Stop the program.

Program:

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
    window_end = min(base + window_size - 1, total_frames - 1)
    print("\n----------")
    print(f"sender window : [Frame {base} - Frame {window_end}]")
    while next_seq < base + window_size and next_seq < total_frames:
        print(f"(Frame {next_seq} Lost in frames+1)")
        next_seq += 1
    if lost_frame is not none and base <= lost_frame < next_seq:
    print(f"Node B --> Frame {lost_Frame} missing:
    Discarding out-of-order frames after it")
```

output:

SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go - Back - N).

Enter total number of frames to send : 8

Enter sender's window size : 4

Enter frame number to simulate as Lost : 5

Sender window : [Frame 0 - Frame 3]

Node A -> Sending Frame 0

Node A -> sending Frame 1

Node A -> sending Frame 2

Node A -> Sending Frame 3

Node B -> Received Frame 0, Sending ACK 0

Node B -> Received Frame 1, sending ACK 1

Node B -> Received Frame 3. Sending ACK 3

Node A -> ACK 3 received, sliding window forw

Sender window : [Frame 4 - Frame 7]

Node A -> Sending Frame 4

Node A -> sending Frame 5

Node A -> sending Frame 6

Node A -> sending Frame 7

Node B -> Frame 5 missing.

Nod A -> timeout. Resending window starting from frame 5.

All 8 frames successfully transmitted fr Node A to Node B

```
print (f" Node A -> Time out ! Resending window
        starting from Frame {lost_frame}.")
next_seq = lost_frame

lost_frame = none
succeds
continue

for i in range (base, next_seq):
    print (f" Node B -> Received frame {i},
    sending ACK {i}")
print (f " Node A -> Ack {next_seq-1} recieved
print (f "All {total_frames} frames successfully
    transmitted from Node A to Node B.")
def main():
    print (" == SLIDING WINDOW PROTOCOL -
                FLOW CONTROL SIMULATION
    total_frames = int (Input ("Enter total number
        of frames to send :")
    lost_frame = int (input (f" Enter Frame
        number to simulate as Lost \n)
    If lost_frame == -1
        lost_frame = None
    simulate (total_frames, window_size, lost_frame)
if __name__ == "__main__":
main()
```

Result: The program to implement flow control at the Datalink layer using the sliding window Protocol was successfully written.